PowerCyber: A Remotely Accessible Testbed for Cyber Physical Security of the Smart Grid

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***Abstract*—Cyber Physical Systems (CPS) security testbeds serve as a platform for evaluating and validating novel CPS security tools and technologies, accelerating the transition of state-of-the-art research to industrial practice. The engineering of CPS security testbeds requires significant investments in money, time and modeling efforts to provide a scalable, high-fidelity, real-time attack-defense platform. Therefore, there is a strong need in academia and industry to create remotely accessible testbeds that support a range of use-cases pertaining to CPS security of the grid, including vulnerability assessments, impact analysis, product testing, attack-defense exercises, and operator training. This paper describes the implementation architecture, and capabilities of a remote access and experimental orchestration framework developed for the PowerCyber CPS security testbed at Iowa State University (ISU). The paper then describes several engineering challenges in the development of such remotely accessible testbeds for Smart Grid CPS security experimentation. Finally, the paper provides a brief case study with some screenshots showing a particular use case scenario on the remote access framework.**

I. Introduction

Several authoritative government reports and technical literature have documented increasing concerns for highly sophisticated, advanced persistent cyber threats on critical infrastructures like the power grid [1]–[4]. Cyber-Physical System (CPS) security testbeds are essential elements in the R&D ecosystem to develop and validate novel CPS security tools and algorithms for securing the grid and making it attack-resilient. However, the engineering of CPS security testbeds requires significant investments in money, time and modeling efforts to provide a scalable, high-fidelity attack-defense platform. Therefore, there is a compelling need to develop remotely accessible testbeds that provide access to the testbed resources for a larger research community, which enables them to perform realistic experimentation on CPS security of the Smart Grid applications.

The main contribution of this paper is the design and development of a remote access and experimental orchestration framework for the PowerCyber testbed at ISU. The remainder of the paper is organized as follows. Section II briefly describes existing CPS security testbed efforts and the use-cases relating to power grid cyber security and also their suitability with respect to providing user access for remote experimentation. Section III presents a quick overview of the current architecture of the PowerCyber testbed at ISU and use-cases that are supported. Section IV elaborates on the implementation architecture of the remote access framework. Section V highlights some of the engineering challenges in enabling remote access for CPS security testbeds. Section VI presents a case study application that was evaluated using the testbed. Section VII presents an overall summary and identifies directions for future work.

II. Previous Work

In the past decade, several educational institutions and national labs have been developing CPS security testbeds for validating and evaluating the various cyber security tools and technologies, and also to provide realistic test environments for attack-defense experimentation. In this section, we identify a few prominent CPS security testbeds along with a brief discussion of the use-cases they support and their capabilities to enable remote experimentation.

The National SCADA Testbed (NSTB) at Idaho National Laboratory (INL) is a one of the first CPS security testbeds in the nation. It consists of physical and cyber system resources to perform vulnerability assessments and impact analysis studies [5]. Several other notable CPS security testbeds have been developed in the recent years, such as the Virtual Control System Environment testbed at Sandia National Laboratory [6], Virtual Power System Testbed at the University of Illinois at Urbana-Champaign [7], powerNET testbed at Pacific Northwest National Laboratory [8], Distribution Cyber Security Testbed at National Renewable Energy Laboratory [9], SCADA security testbed at Mississippi State University [10], and SCADA testbed at University College, Dublin [11].

The DETER testbed at University of Southern California's Information Sciences Institute, is a large-scale testbed for cyber security experimentation [12]. DETER is a multi-user testbed that provides remote users the tools to create custom experiments by provisioning appropriate computing resources such as virtual machines on the dedicated server clusters. However, the DETER testbed lacks CPS resources, specific to the power grid such as Supervisory Control And Data Acquisition (SCADA) software, Remote Terminal Units (RTU), relays and phasor measurement units, to perform hardware-in-the-loop experimentation in conjunction with its computing resources.

Some of the common use-case scenarios that these above mentioned CPS security testbeds support are: vulnerability assessments on SCADA devices communication protocols, and software, impact analysis for realistic cyber attack scenarios, developing defense measures such as anomaly/intrusion detection, and security product testing, etc,. However, the real utility of the CPS security testbeds for the academic and

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industry research community at large has been limited by the extent of remote access to these testbeds. Therefore, there is a compelling need to develop and maintain a remotely accessible CPS testbed to a broad user community. Such a testbed would enable a large user base to perform realistic, high fidelity, hardware-in-the-loop (HIL), attack-defense experimentation for improving the security and resiliency of the Smart Grid.

III. PowerCyber CPS Security Testbed

A. Architecture

Figure 1 shows the current architecture of the PowerCyber CPS Security testbed at ISU. PowerCyber consists of a hybrid mix of industry standard hardware and software, emulated components and real-time power system simulators for real-time, hardware-in-the-loop CPS security experimentation [13], [14]. Specifically, the testbed consists of SCADA hardware/software from Siemens that include substation automation system (SICAM PAS), control center software (Power TG), SCADA and substation communication protocols (DNP3, IEC 61850, IEEE C37.118), and security technologies (Scalance: Firewall, VPN), four multifunction protection relays (7SJ610, 7SJ82), three SEL 421 Phasor Measurement Units (PMU) and a Phasor Data Concentrator (PDC).

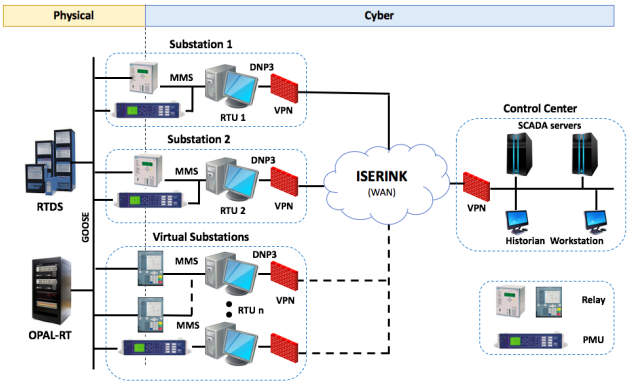

Fig. 1. PowerCyber CPS Security Testbed Architecture

B. Use cases Supported

PowerCyber has been leveraged to support the following types of use cases that are relevant to Cyber Physical security experimentation for the power grid.

1) Vulnerability Assessment: PowerCyber supports vulnerability assessment of industry grade SCADA software and hardware platforms, network protocols and configurations to detect unknown vulnerabilities based on standard vulnerability databases. Several unknown vulnerabilities were discovered in the past and a responsible disclosure process was followed to disclose them to vendors and regulatory authorities [15].

2) Impact Analysis & Mitigation Research: PowerCyber provides a realistic virtual critical infrastructure environment wherein realistic experiments on wide-area monitoring, protection, and control in the smart grid environment can be carried out. Specifically, the testbed has been used to evaluate coordinated attacks on Remedial Action Schemes [13], and data integrity attacks on Automatic Generation Control [16]. Also, the testbed allows the implementation of defense measures such as firewalls, intrusion detection systems, software patches, etc., to also evaluate the performance of various mitigation strategies against attack vectors.

3) Testbed Federation: Recently, the PowerCyber testbed was successfully federated with the DETER testbed and the ISERINK platform, as part of Smart America Challenge and NIST Global City Teams Challenge to create a large-scale, high-fidelity CPS security testbed environment [17], [18]. The federated testbed was used to demonstrate proof-of-concept attack-defense experimentation on a wide-area protection scheme. Additional use cases to showcase the utility of CPS testbed federation are currently being explored.

4) Attack-Defense Exercises & Operator training: The PowerCyber testbed and the ISERINK platform for cyber defense competitions [19] had been recently integrated to conduct realistic cyber attack-defense training exercises for utility practitioners at the NERC GridSecCon 2015 conference [20]. Also, PowerCyber has been used every year as part of an industry workshop to provide hands-on training sessions.

5) Model Development: PowerCyber testbed is currently developing a repository of standardized models and experimental datasets for power systems and associated cyber systems to facilitate researchers in leveraging the testbed capabilities for cyber security experimentation.

6) Remote Access: In order to engage and enable a broad user community to perform a variety of power grid related cyber security experimentation, a remote access framework is currently being developed on the PowerCyber testbed. Specifically, the remainder of this paper will describe the implementation architecture of a remote access framework for the PowerCyber testbed, and also identify some important challenges in sustaining a remotely accessible testbed.

IV. PowerCyber: Remote Access Framework

Figure 2 shows the implementation architecture of the remote access framework for the PowerCyber testbed. The architecture describes a remote access CPS security testbed that consists of several interconnected VM's (SCADA, power system control, attack, defense, gateway), power system simulators, and physical devices such as relays and PMU's for performing hardware-in-the-loop experimentation.

The remote access framework interacts with the remote users via a Front-end Web server, which provides a web-based user interface that allows users to configure the relevant parameters needed for the automated attack-defense experimentation. The front-end web server is integrated tightly to a backend orchestration framework, which is written in Python. The backend framework orchestrates the creation/configuration of the necessary cyber resources such as the SCADA virtual machines, manages the interaction with the power system simulators and physical devices such as the relays and PMUs, coordinates various attack actions and defense measures depending on the user input, and also provides a way for the users to collect relevant simulation artifacts from the cyber

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system and the physical system in the form of log files, packet captures, plots, etc.

The power system configuration module performs actions such as model compilation, loading, and execution on a real-time power system simulator and appropriately maps components such as relays and PMUs on the power system model based on user selection. The SCADA configuration module spawns SCADA virtual machines such as the control center VM's that house the Energy Management Systems (EMS) software, substation RTU VM's, and connects these VM's according to user specified communication topology. The module also appropriately initializes SCADA communications between the various devices and verifies communications between the SCADA VM's and the hardware devices that are mapped into the experiment. Based on the user's selection, the attack module coordinates and triggers attack actions on appropriate SCADA components. The attack module spawns attacker virtual machines once all the initializations have been performed appropriately. The attack script includes user selected attack actions from the attack template library. The defense module implements host based or network based mitigation mechanisms such as firewall rules, intrusion/anomaly detection systems, and/or moving target defense (MTD) appropriately based on user input. Once the experiments are performed, the backend framework can pull cyber system artifacts such as firewall and intrusion detection log files, network packet captures, network performance statistics, etc., from the various SCADA VM's and the gateway/defense nodes. Also, as mentioned previously, the backend interacts with the power system control scripts to pull relevant physical system simulation artifacts.

Depending on the experimental scenario chosen by the user, appropriate inputs are obtained for the scenario with respect to the power system model, hardware device mapping, communication network topologies, attack parameters and defense parameters. This allows the users to run their own experiments within the limitations of the template experimental scenarios that are provided as initial inputs. As the capabilities of the remote access framework mature, more flexibility would be available to the users to select their own experimental scenarios by using custom power system models, relay/ PMU configurations, attack vectors, and defense mechanisms such as anomaly/intrusion detection algorithms. The following sections describe the design flow of the remote access framework.

A. Design Flow for Remote, Automated Experimentation

The key steps in the overall design flow for performing automated, cyber attack/defense experimentation are: (1) Power system configuration, (2) WAMPAC experiment selection, (3) Cyber system configuration, (4) Defense configuration, (5) Attack Configuration, (6) Collecting Cyber System Results, (7) Collecting Physical System Results. These overall design flow steps are described in detail with respect to user interface and experiment automation below.

1) User Interface: The user interface allows the user to select the power system model, the WAMPAC experimental

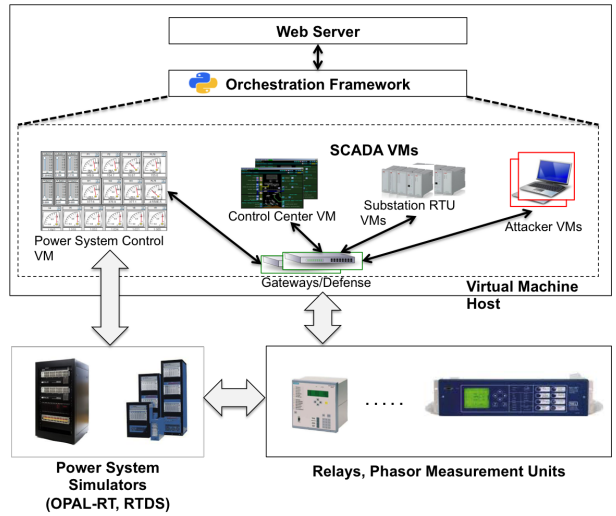

Fig. 2. Remote Access Framework - Implementation Architecture

scenario, and also select attack type and targets. As part of defense measures, the user can select from either host based defense or network based defense. Based on their inputs, user interface provides options to select from defense methodologies such as firewall, intrusion detection and prevention based defense, etc., for implementation. For orchestrating the attack, the user interface provides a list of attack vectors such as DOS attack, Command injection attack, coordinated attack to choose from, along with options to choose the attack locations on the system selected. Once the experiment has been performed, the user interface facilitates the collection of cyber system impacts such as downloading packet captures, log files etc., during attack phase. As part of power system impacts, the user interface provides the relevant data collection options for plotting power system parameters.

2) Experiment Automation: With respect to the power system configuration, the automation tasks involve loading, compilation of the appropriate power system model such as the IEEE 9 bus system model on the simulator, initializing the runtime interface, and ensuring that the model has reached its steady state operating condition before any attack actions are started. Depending on the WAMPAC experiment selection made, the backend automation scripts would configure physical components accordingly so that they could be mapped into the power system model for hardware-in-the-loop experiments, and also verify their integration with the runtime interface.

With respect to automating the cyber system, the backend configuration module spawns control center and substation VM's appropriately, and initializes the VM's in a specific network topology. Also, the scripts verify if inter-device communication is successful before proceeding further. Based on user selection, the backend scripts implement and initialize network based or host based defense measures such as firewalls, or intrusion detection systems on gateway nodes or on the hosts such as substation VM's. As part of attack orchestration, the backend scripts would spawn attacker VM's, which have a library of pre-defined attack modules. Based on the experiment

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scenario, the master script coordinates and triggers the specific attack scripts on these VM's. As part of collecting the results on the cyber layer, the backend scripts would pull log files from the VM's where defense measures such as firewall and IDS logs were deployed, and packet captures from the gateway node, besides other network performance statistics. Similarly, the scripts on the power system control node would pull power system simulation artifacts such as data files or plots and make them available to the user as experimental results.

V. Challenges For Remote access

The unique nature of CPS testbeds with its complex cyber-physical interdependencies, and an inherent diversity of simulated, emulated and physical components brings several practical engineering challenges in order to create a scalable, flexible, multi-user, and remotely accessible testbed. This section will describe some of the important engineering challenges in creating a scalable, flexible, and remotely accessible testbed for CPS security experimentation.

Scalability: One of the most critical challenges in creating remote access testbeds is scalability. As we increase the scale of the power system model, the associated cyber system model increases. This translates to increased resource requirements for performing real-time simulations with hardware-in-the-loop experimentation. Therefore, power system scalability would be limited to the maximum system size that the power system simulator can support for a real-time simulation. Scalability in cyber system virtual machines (VMs) could be easily overcome with deploying additional computing resources such as server racks, however, increasing power system simulation capabilities at the same level is prohibitively expensive. To some extent, this could be handled by abstracting and equivalencing parts of the system model. Though scalability for physical components like relays and PMU's also comes at a price, to some extent we can get around device scalability with device emulators, which replicate the entire communication behaviors that are essential to cyber security experimentation.

Concurrency and Isolation: One of the main differences between the implementation of a remote access framework for a CPS security testbed vs. a regular cyber security testbed would be the problem of concurrent, multi-user experiments. There exist several options to virtualize and compartmentalize computing resources like VM's to provide isolation, security for multiple-concurrent users. However, the problem of sharing expensive, limited physical components such as relays, PMU's, and real-time power system simulators between multiple concurrent users is much more complicated and requires additional research and development effort.

Flexibility: Extending the discussion on concurrency, the flexibility of a CPS security testbed is much lesser when compared to a traditional cyber security testbed. Specifically, the configuration of physical components such as relays and PMU's need to be changed dynamically based on user inputs for different types of experiments. Some of the software that are used to program these devices are based on a graphical user interface and therefore are not very amenable to automated scripting. While there are command line tools that automate mouse clicks on GUI, this approach is not very flexible and suitable for a wide range of scenarios.

Troubleshooting: Troubleshooting remote, automated experiments when things don't work as expected presents tough challenges on any type of testbed. Especially, in CPS environments, with SCADA devices, and communication protocols, this challenge is even harder to overcome due to inherent design weaknesses in terms of security, and the heterogeneity of devices. Typically, SCADA devices have been built with usability, reliability as a preference over security. This leads to unexpected failure modes in devices during cyber attack-defense experimentation. Consequently, failures are hard to debug in an automated manner. Therefore, the remote access provided to users would be mostly driven based on experiment templates, rather than complete customization.

VI. Case Study

This section briefly showcases how PowerCyber's remote access framework was used to perform a *coordinated attack/defense experiment on a wide-area protection scheme*, as specified in [13], and presents a few screenshots (Figure 3, and Figure 4). The users can configure power system, cyber system, attack and defense actions as part of their experiment as shown in Figure 3.a. For this case study, the IEEE 9-bus power system model was chosen, and RAS scheme was mapped on a specific location in that model (Figure 3.b). The web interface enables the user to execute different types of attack options (Figure 4.a), and observe/collect power system impacts from runtime interface of the simulator (Figure 4.b).

VII. Conclusions and Future work

The need for remotely accessible testbeds to enable CPS security experimentation for the power grid is quite compelling. This paper describes the implementation architecture, and capabilities of PowerCyber's remote access and experimental orchestration framework for CPS security experimentation on Smart Grid applications. The remote access implementation architecture described in this paper is actively being refined, nevertheless, it provides a solid foundation for further R&D in enabling remote experimentation. The paper also articulates various challenges at large that need to be addressed in creating and sustaining remotely accessible CPS security testbeds.

Future work: The following activities would be performed: (i) developing a library of models, and datasets, (ii) developing a larger user community, (iii) improving programmability/customization, and (iv) developing advanced use cases.

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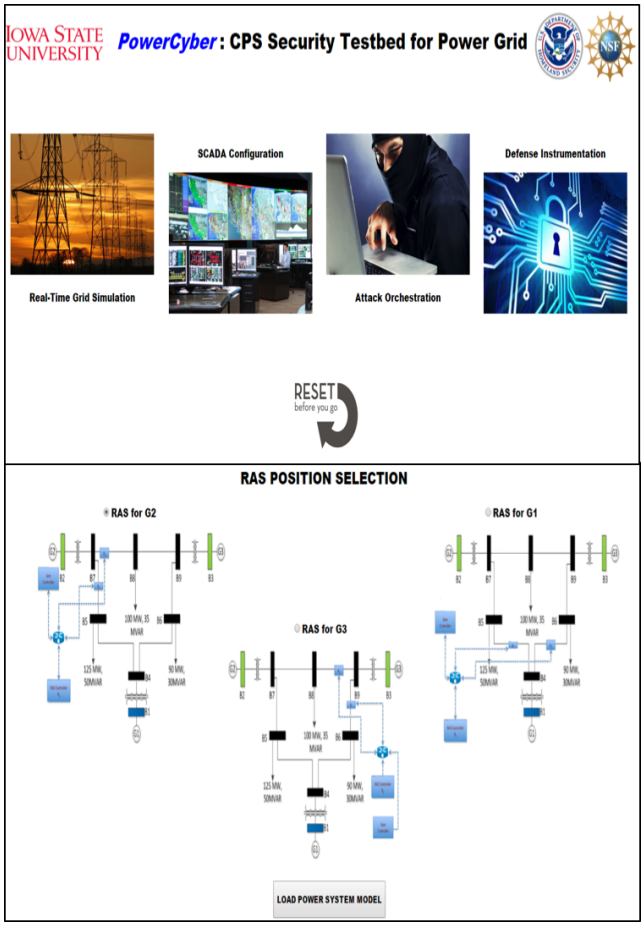

Fig. 3. Case Study - (a) Experiment Configuration (top), (b) RAS position

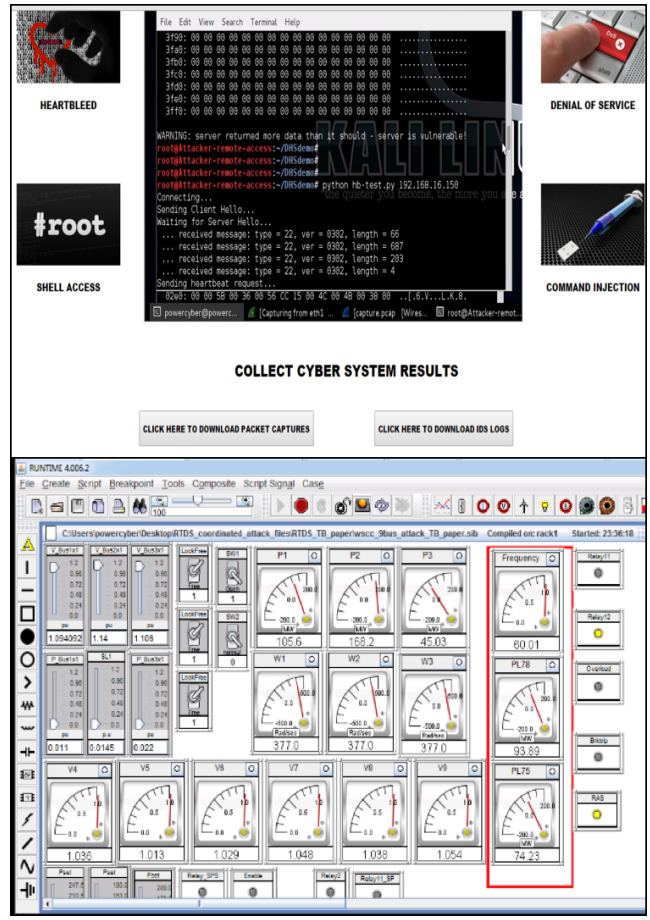

Fig. 4. Case Study - (a) Attack actions (top), (b) Power System Output

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